# REGIMES OF PEARLING DYNAMICS IN TUBULAR VESICLES

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## ABSTRACT

Pearling converts membrane tubes into beads joined by thin tethers. We combine linear stability analysis and axisymmetric simulations to study pearling under different stretching protocols.

In an extensional flow, marginal initial perturbations make pearls form first near the vesicle centre, as observed by Kantsler et al. [1], rather than through the end-to-centre front found by Narsimhan et al. [2].

Above the critical strain rate for loss of metastability in an extensional flow [3], unbounded elongation rapidly increases tension until pearling becomes inevitable. Continued stretching of the connecting tethers causes secondary instabilities and a cascade of progressively smaller beads.

By contrast, an approximately constant end force can balance Helfrich elasticity, creating a quasi-stationary chain of long-lived pearls. Individual "free pearls" are soliton-like solutions of the stationary force-balance equation. Their equilibrium tension decreases with radius, driving slow fluid transfer from smaller to larger pearls.

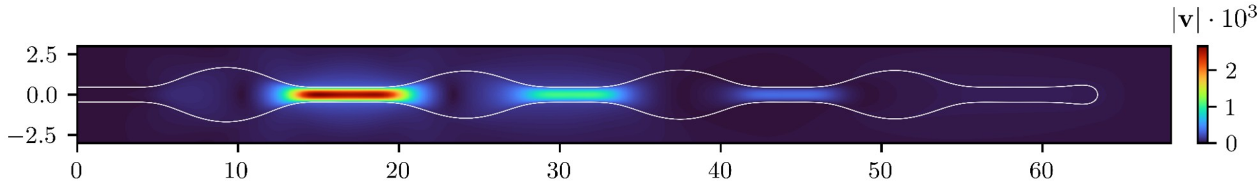


***Fig.*** *Flow field during slow coarsening of a quasi-stationary pearl chain. Colour shows fluid-speed magnitude; pressure gradients drive fluid through thin tethers from smaller to larger pearls.*

## REFERENCES

1 *V. Kantsler, et al., Phys. Rev. Lett., **101**, 048101, (2008), DOI: https://doi.org/10.1103/PhysRevLett.101.048101.*

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